

Nsight example #3

Data Collected: Example #3 collects all the data listed below.

Example #1 output includes:

Field	Description
epoch	Current epoch number
step	Global step counter
loss	Per-step MSE loss (rank 0)
lr	Current learning rate
samples_per_sec	Throughput across all ranks
gpu_mem_gb	GPU memory allocated (rank 0)
epoch_avg_loss	All-reduced mean loss for the epoch
epoch_time	Wall seconds for the epoch

Added in Example #2 output:

Field	Description
grad_norm	Gradient L2 norm before clipping
gpu_mem_reserved_gb	Memory reserved by allocator (allocated + fragmentation headroom)
ms_h2d	Mean ms for host → device transfer, all-reduced across ranks
ms_forward	Mean ms for forward pass, all-reduced across ranks
ms_backward	Mean ms for backward pass + DDP all-reduce, all-reduced across ranks
ms_optim	Mean ms for unscale + clip + optimizer step, all-reduced across ranks
run_tag	Optional string label for multi-run comparison

Added in Example #3 output:

Field	Description
gpu_util_pct	SM utilization % from nvidia-smi driver query
gpu_mem_used_gb	Driver-reported GPU memory in use (GB)
gpu_mem_total_gb	Total physical GPU memory (GB)
ms_dataload	Mean ms for DataLoader <code>__next__()</code> , all-reduced across ranks
ms_nccl	Mean ms for explicit loss all-reduce collective, all-reduced across ranks
achieved_tflops	Estimated TFLOP/s this step (fwd+bwd FLOPs ÷ step wall time)
roofline_pct	Achieved TFLOP/s as % of H100 theoretical peak for the active dtype
epoch_wall_s	Total wall-clock seconds for the epoch
epoch_s_dataload	Epoch total seconds spent in data loading
epoch_s_h2d	Epoch total seconds spent in host → device transfer
epoch_s_forward	Epoch total seconds spent in forward pass

Field	Description
epoch_s_backward	Epoch total seconds spent in backward pass
epoch_s_nccl	Epoch total seconds spent in NCCL collectives
epoch_s_optim	Epoch total seconds spent in optimizer step
epoch_s_idle	Epoch wall time not captured by any instrumented phase
epoch_pct_*	Percentage equivalents of each epoch_s_* field
rank	Rank index of the process that wrote this record (per-rank JSONL only)

Install, Run, and View Results

*First you need to acquire the file `NsightSystems-linux-cli-public-2026.3.1.157-3804839.deb`. This can be found here: <https://developer.nvidia.com/nsight-systems/get-started>

How to create and analyze Nsight outputs from python jobs run via slurm on Ursa GPU nodes

Regular Text, [Commands](#), [Command Return](#)

- A. Copy `/scratch3/SYSADMIN/nescmngmt/Ron.Millikan/share/peter/nsite-files-3.tgz` to your scratch directory in Ursa
- B. Unzip file `tar -xvzf nsite-files-3.tgz`
- C. cd into the `nsite-files/` directory
- D. Create a virtual environment `python -m venv nsite-files-env`
- E. Activate virtual environment `source nsite-files-env/bin/activate`
- F. Install files with pip

```
pip install --upgrade pip
```

```
pip install -r requirements.txt
```

- G. Make Nvidia nsys available for your use

```
ar x NsightSystems-linux-cli-public-2026.3.1.157-3804839.deb
```

```
tar xf data.tar.gz
```

These commands, the first is ‘ar’, not misspelled ‘tar’, will create a directory tree inside your `nsite-files` directory called `opt`.

- H. Test your nsys install `opt/nvidia/nsight-systems-cli/2026.3.1/bin/nsys --version`

Should return *NVIDIA Nsight Systems version 2026.3.1.157-263138048394v0*

- I. Get the exact location of your new nsys binary

```
find . -name "nsys" -type f
```

Should return something like this: *./opt/nvidia/nsight-systems-cli/2026.3.1/target-linux-x64/nsys*

- J. Add nsys to your path for ease of use

Edit `~/bashrc`

```
export PATH= /scratch3/<YOUR DIRECTORY PATH>/nsite_files/opt/nvidia/nsight-systems-cli/2026.3.1/bin:$PATH
```

```
source ~/.bashrc
```

- K. Install Nvidia gui front end on your laptop

<https://developer.nvidia.com/nsight-systems>

- L. Modify your run.sh file. This is a slurm submission script. It needs your account, qos, and partition information.

- M. While in the run.sh file, set the line starting with ‘source’ to your source location, and the line starting with ‘SCRIPT’ to the correct location of your gemm_nsight.py file.

- N. Run it – `sbatch run.sh`

- O. Once run is complete it is recommended that you tar or zip the contents of the logs/ and profiles/ directories and transfer this file to your local machine. Untar the file.

- P. Open the Nvidia gui app. Open the .sys-rep file. Enjoy.

Notes – This code runs a ‘warm-up’ process prior to collecting data. If you need shorter run times consider commenting out this part of the code.